

9-5

Polynomial space $\mathbb{P}_n$ (Axler 2.A)

Def 2.14 $\mathbb{P}_n(\mathbb{R}) = \left\{ \sum_{i=0}^n a_i x^i : a_i \in \mathbb{R} \right\}$

$\mathbb{P}_n(\mathbb{R})$ is a vector space of dim $n+1$ with standard basis $\{1, x, x^2, \dots, x^n\}$ $e_j = x^{j-1}$

We can turn $\mathbb{P}_n(\mathbb{R})$ into an inner product space by $\langle \cdot, \cdot \rangle : \mathbb{P}_n \times \mathbb{P}_n \rightarrow \mathbb{R}$ defined as

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx$$

This admits the L^2 norm (over $\mathbb{R}$)

$$\|f\|_2 = \sqrt{\langle f, f \rangle} = \left(\int_0^1 f(x)^2 dx \right)^{1/2}$$

General inner product $\langle f, g \rangle_{w(x)} = \int_{-\infty}^{\infty} f(x)g(x)w(x)dx$

Exercise

- 1) Show that the following subsets of $\mathbb{P}_3(\mathbb{R})$ are subspaces and find bases

$$a. \mathcal{U}_1 = \{ f \in \mathcal{P}_3(\mathbb{R}) : f(1) = 0 \}$$

$$b. \mathcal{U}_2 = \{ f \in \mathcal{P}_3(\mathbb{R}) : \langle f, x \rangle = 0 \}$$

Solutions:

$$\begin{aligned} a. \text{ Clearly } 0 \in \mathcal{U}_1. \text{ Now let } f_1, f_2 \in \mathcal{U}_1, \\ \text{so that } f_1(1) = f_2(1) = 0. \text{ Now as} \\ (\lambda_1 f_1 + \lambda_2 f_2)(1) = \lambda_1 f_1(1) + \lambda_2 f_2(1) \\ = \lambda_1 \cdot 0 + \lambda_2 \cdot 0 = 0 \end{aligned}$$

we have that $\mathcal{U}_1$ is a subspace.

$$\text{letting } f = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \in \mathcal{U}_1$$

$$\text{we have that } a_0 + a_1 + a_2 + a_3 = 0.$$

Now $a_0 = -(a_1 + a_2 + a_3)$ and thus
the general solution to the coeff equation
is given by

$$\begin{aligned} \vec{a} &= \begin{bmatrix} -a_1 - a_2 - a_3 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} \\ &= a_1 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} 1 \\ x \\ x^2 \\ x^3 \end{matrix} + a_2 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

Thus a basis for $\mathcal{U}_1$ is given by

$$\{x-1, x^2-1, x^3-1\}$$

Another bases is $\{x-1, x^2-x, x^3-x^2\}$

b. Proving that U_2 is a subspace is similar so we skip ahead to finding a basis. let $f \in U_2$ s.t. $f = a_0 + a_1x + a_2x^2 + a_3x^3$. Then as

$$\begin{aligned} 0 &= \langle f, x \rangle = \int_0^1 x \cdot f(x) dx \\ &= \int_0^1 a_0x + a_1x^2 + a_2x^3 + a_3x^4 \\ &= \frac{a_0}{2} + \frac{a_1}{3} + \frac{a_2}{4} + \frac{a_3}{5} \end{aligned}$$

and thus $30a_0 + 20a_1 + 15a_2 + 12a_3 = 0$

Now as $a_0 = \frac{-20a_1 - 15a_2 - 12a_3}{30}$

we obtain

$$\vec{a} = \begin{bmatrix} -\frac{2a_1}{3} - \frac{a_2}{2} - \frac{2a_3}{5} \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$= a_1 \begin{bmatrix} -2/3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} -1/2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} -2/5 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

and thus a basis for U_2 is

$$\left\{ x - \frac{2}{3}, x^2 - \frac{1}{2}, x^3 - \frac{2}{5} \right\}$$

Under the identification

$$\sum_{i=0}^n a_i x^i \rightarrow \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

we have that $\mathbb{P}_n(\mathbb{R}) \cong \mathbb{R}^{n+1}$ (isomorphic)

However when we compute

$$\begin{aligned} \langle a_0 + a_1 x, b_0 + b_1 x \rangle &= \int_0^1 a_0 b_0 + (a_1 b_0 + a_0 b_1) x + a_1 b_1 x^2 \\ &= a_0 b_0 + \frac{(a_1 b_0 + a_0 b_1)}{2} + \frac{a_1 b_1}{3} \end{aligned}$$

we don't get $[a_0 \ a_1] \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = a_0 b_0 + a_1 b_1$

The reason is that the standard basis for $\mathbb{P}_n(\mathbb{R})$ is not orthonormal.

Complex Vector Space $\mathbb{C}^n$

Def 3.1 The vector space $\mathbb{C}^n$ (over $\mathbb{C}$) consists of n -tuples of $\mathbb{C}$

$$\mathbb{C}^n = \{ (z_1, \dots, z_n) : z_i \in \mathbb{C} \}$$

Note that $\mathbb{R}^n \subseteq \mathbb{C}^n$ but not a subspace.
The standard basis for $\mathbb{C}^n$ is the same as $\mathbb{R}^n$.

Inner Product and Norm $\overline{x+iy} = x-iy$

$$i. \quad \langle \vec{z}, \vec{w} \rangle_{\mathbb{C}} = \sum_{i=1}^n z_i \bar{w}_i = \vec{w}^* \vec{z}$$

where $\vec{w}^* = (\vec{w}^T)^{\text{conj.}}$ (conj. transpose)

$$ii. \quad \|\vec{z}\|_{\mathbb{C}} = \sqrt{\langle \vec{z}, \vec{z} \rangle_{\mathbb{C}}} = \left(\sum_{i=1}^n |z_i|^2 \right)^{1/2}$$

$$\text{color: red; } z_i \bar{z}_i = |z_i|^2$$

$$\vec{w}^* = \vec{w}^{\dagger} = \vec{w}^H$$

(Axler §3, Strang §8)

Linear maps of Vector Spaces

Def 3.2 Let V and W be vector spaces over

F . Then $T \in \mathcal{L}(V, W)$, i.e. T is a linear map

from V to W iff for any $\vec{v}_1, \vec{v}_2 \in V$ and $\lambda_1, \lambda_2 \in F$ we have that

$$T(\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2) = \lambda_1 T(\vec{v}_1) + \lambda_2 T(\vec{v}_2)$$

Examples

- i. Inner product : $T_{\vec{v}} : \mathbb{R}^n \rightarrow \mathbb{R}$ by $T_{\vec{v}} \vec{x} = \langle \vec{x}, \vec{v} \rangle$
- ii. Cross product : $T_{\vec{v}} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T_{\vec{v}} \vec{x} = \vec{x} \times \vec{v}$
- iii. Linear Diff operator : $L : \mathcal{P}_n \rightarrow \mathcal{P}_n$ by
$$L(f) = a_0 f + a_1 f' + \dots + a_n f^{(n)}$$

Theorem 3.3 A linear map is uniquely determined by its action on a basis.

Proof: Let $S, T \in \mathcal{L}(V, W)$ and let $\{\vec{v}_1, \dots, \vec{v}_n\}$ be a basis for V . Now if $S\vec{v}_i = T\vec{v}_i$ for each i , then for any $\vec{x} \in V$ we have that $\vec{x} = \sum_{i=1}^n \lambda_i \vec{v}_i$ and thus

$$\begin{aligned}(S-T)(\vec{x}) &= (S-T)\left(\sum \lambda_i \vec{v}_i\right) \\ &= \sum \lambda_i (S-T)(\vec{v}_i) \\ &= \sum \lambda_i \vec{0} = \vec{0}\end{aligned}$$

Hence $S-T=0$ (0-map) and $S=T$.

Exercise

2) let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ be defined by

$$T\vec{e}_1 = 3\vec{e}_1 - 2\vec{e}_2, \quad T\vec{e}_2 = \vec{e}_1 + 6\vec{e}_2$$

Compute $T\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ and $T\vec{x}$ where $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$\begin{aligned} T\begin{bmatrix} 2 \\ 4 \end{bmatrix} &= T(2\vec{e}_1 + 4\vec{e}_2) = 2T\vec{e}_1 + 4T\vec{e}_2 \\ &= 10\vec{e}_1 + 20\vec{e}_2 = \begin{bmatrix} 10 \\ 20 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} T\vec{x} &= T(x_1\vec{e}_1 + x_2\vec{e}_2) = x_1T\vec{e}_1 + x_2T\vec{e}_2 \\ &= x_1\begin{bmatrix} 3 \\ -2 \end{bmatrix} + x_2\begin{bmatrix} 1 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 1 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

Note that $A = \begin{bmatrix} 3 & 1 \\ -2 & 6 \end{bmatrix}$ is the "matrix representation" of T w.r.t. the standard basis.

Isomorphism

Def 3.4 let $T \in \mathcal{L}(V, W)$. We say that T is injective if $T\vec{x}_1 \neq T\vec{x}_2$ whenever $\vec{x}_1 \neq \vec{x}_2$ and (one-to-one)

surjective if for every $\vec{y} \in W$, there exists $\vec{x} \in V$
(onto) If T is surj. then $\text{range}(T) = W$

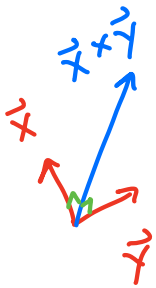
s.t. $\vec{y} = T\vec{x}$. If T is injective and surjective
then it is bijective.

Def 3.5 If $T \in \mathcal{L}(V, W)$ is a bijective then
we say that T is an isomorphism from V to W
and write $V \cong W$ (V is isomorphic to W)

Example: An isomorphism of $\mathbb{P}_n(\mathbb{R})$ to $\mathbb{R}^{n+1}$ is
given by $T(x^i) = \vec{e}_{i+1}$

Exercise

3) Show that for non zero $\vec{v} \in \mathbb{R}^3$, the
linear map $T_{\vec{v}} \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^3)$ defined by
 $T_{\vec{v}}(\vec{x}) = \vec{x} \times \vec{v}$ is neither injective nor
surjective.



T is not injective as $T_{\vec{v}}(\vec{0}) = T_{\vec{v}}(\vec{v}) = \vec{0}$
(more generally $T_{\vec{v}}(\lambda\vec{v}) = \vec{0}$ for any λ)

T is not surjective as $T_{\vec{v}}(\vec{x})$ is
orthogonal to $\vec{v}$ for all $\vec{x}$ and $\vec{v} \neq \vec{0}$.

$$(T_{\vec{v}}(\vec{x}) \neq \vec{v})$$

Kernel and Range

Def 3.6 let $T \in \mathcal{L}(V, W)$. We define the kernel of T as the set $\ker T = \{ \vec{x} \in V : T\vec{x} = \vec{0}_W \}$ and the range of T as the set $\text{range } T = \{ \vec{y} \in W : \vec{y} = T\vec{x} \text{ for some } \vec{x} \in V \}$
(Image of T)

Theorem 3.7 $\ker T$ and $\text{range } T$ are subspaces of V and W respectively.

Theorem 3.8 let $\dim V = \dim W = n$. Then $T \in \mathcal{L}(V, W)$ is an isomorphism iff $\ker T = \emptyset$.
I.e. T is inj and surj. iff $\ker T = \emptyset$.

Def 3.9 If $T \in \mathcal{L}(V, W)$ is an isomorphism then there exists inverse $T^{-1} \in \mathcal{L}(W, V)$ s.t.

$$T \circ T^{-1} = \text{id}_W \quad \text{and} \quad T^{-1} \circ T = \text{id}_V$$

$$\text{id}_V(\vec{x}) = \vec{x} \quad \text{for all } \vec{x} \in V$$

Exercise

4) Let $T \in \mathcal{L}(V, W)$ be such that

$T\vec{v}_i = \vec{w}_i$ for $i=1, \dots, n$ where

$\{\vec{w}_1, \dots, \vec{w}_n\}$ are linearly independent. Show

that $\{\vec{v}_1, \dots, \vec{v}_n\}$ must be linearly independent as well.

Assume for contradiction that $\vec{v}_j = \sum_{i \neq j} c_i \vec{v}_i$

Now applying T we have that

$$\vec{w}_j = T\left(\sum_{i \neq j} c_i \vec{v}_i\right) = \sum_{i \neq j} c_i \vec{w}_i$$

which contradicts the lin. ind. of the $\vec{w}_i$'s.